

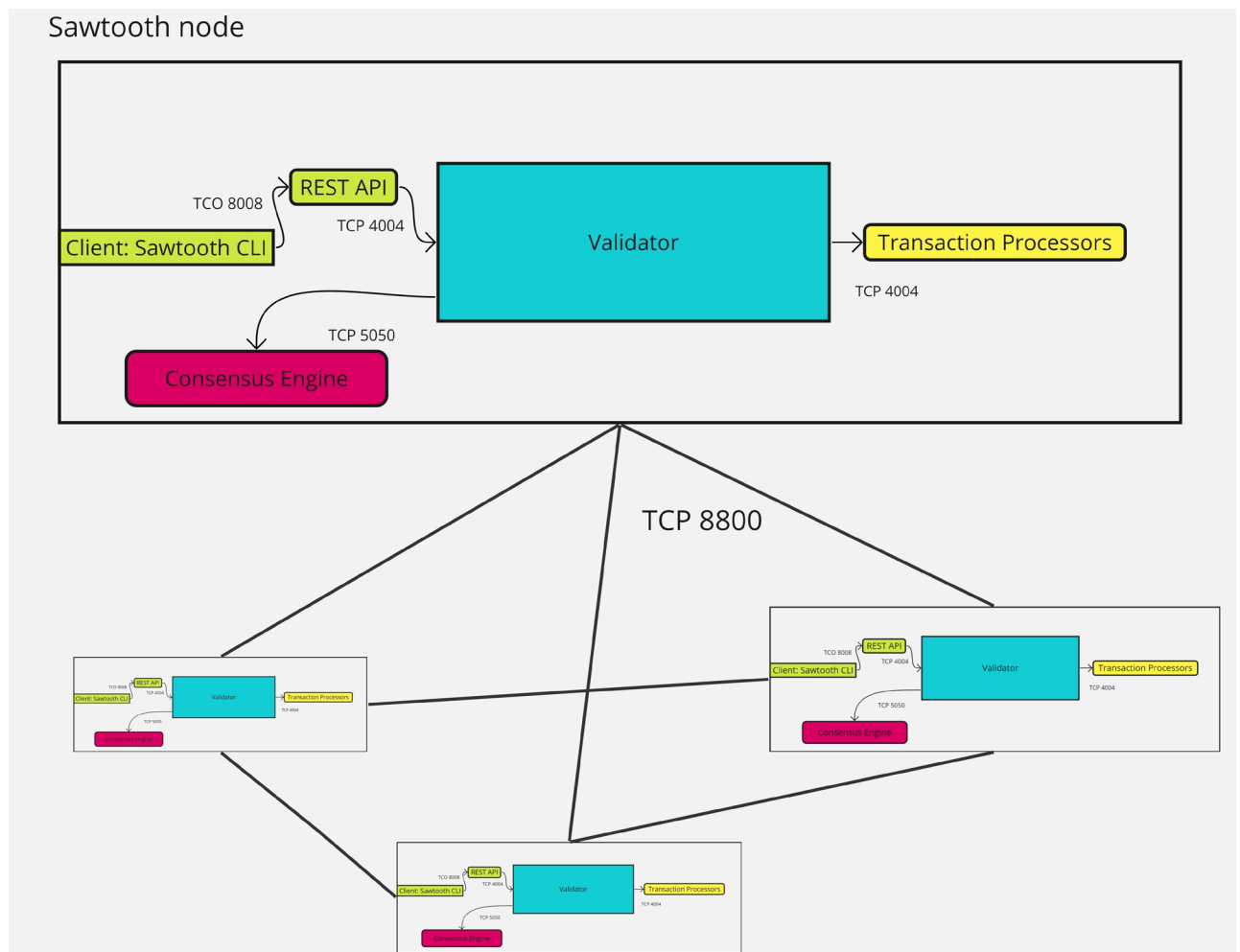


Figure 1: Sawtooth node and network

periphery of the blockchain. This freedom fosters innovation and streamlines the development process without compromising the core functionalities of the ledger.

Components of Sawtooth nodes

Each node in this Sawtooth network runs a validator, a REST API, a consensus engine, and transaction processors. All the components are depicted in Figure 1.

- **Clients:** These applications, representing the user interface, interact with the validator nodes using a Representational State Transfer (RESTful) API. This API facilitates the submission of transactions and retrieval of data from the ledger.
- **RESTful API:** This user-friendly interface acts as a bridge between clients and validators. It provides a language-independent mechanism for applications to communicate with the validators, promoting interoperability and simplifying integration.
- **Transaction processors:** These intelligent contracts serve as the backbone of business logic enforcement within Sawtooth. They meticulously validate transactions, ensuring adherence to predefined rules before approving their inclusion in the ledger. Transaction processors act as the gatekeepers, safeguarding the integrity of the shared record.
- **Validator:** This critical component shoulders the responsibility of verifying the validity of both transactions and blocks. Validators meticulously scrutinise each unit of data before committing them to the tamper-proof ledger. By acting as the gatekeeper, validators ensure the consistency and reliability of the shared record.
- **Consensus engine:** This mechanism serves as the foundation for an agreement among all validators regarding the current state of the ledger. Hyperledger Sawtooth boasts a modular design philosophy that extends to the consensus engine as well. It offers a variety of pluggable consensus algorithms, including